MIL-I-21070B(SHIPS)
2 August 1965

SUPERSEDING
MIL-I-21070A(SHIPS)
31 July 1959
(See 6.5)

MILITARY SPECIFICATION

INSULATING SHEET AND TAPE, ELECTRICAL, REINFORCED MICA PAPER

1. SCOPE

1.1 Scope.- This specification covers flexible reinforced mica paper electrical insulation used in the coils and slots of electrical machinery.

1.2 Classification.-

1.2.1 Classes.- The electrical insulation shall be of the following classes, as specified (see 6.2):

Class 130* - With organic resin binder and organic varnish coating (if used) on glass fiber woven fabric reinforcement.
Class 155* - With epoxy or polyester resin binder and an epoxy or polyester coating (if used) on glass fiber woven fabric reinforcement.
Class 180 - With silicone resin binder and silicone varnish coating (if used) on glass fiber woven fabric reinforcement.
*Binder and coating must be free of any traces of silicones.

1.2.2 Forms.- The electrical insulation shall be of the following forms, as specified (see 6.2):

Form S - Flat Sheets (see 3.2.1).
Rolls 4 to 40 inches wide (see 3.2.1).
Form T - Tape in rolls up to 4 inches wide (see 3.2.2).

1.2.3 Types.- The electrical insulation shall be of the following types conforming to the construction and thicknesses specified in table I, as specified (see 6.2):

Table I - Construction and thickness.

Type	Thickness of composite			Thickness for each component 1/		
	Inch			Inch		
	Nominal	Maximum average	Minimum average	First layer suggested minimum	Second layer minimum average	Third layer suggested minimum
F mp F - 5	0.005	0.006	0.0045	0.001 F	0.002 mp	0.001 F
F mp F - 10	.010	.011	.009	.001 F	.006 mp	.001 F
F mp F - 15	.015	.0165	.014	.003 F	.007 mp	.003 F
G mp F - 6	.006	.007	.005	.002 G	.002 mp	.00025 F
G mp F - 8	.008	.009	.007	.002 G	.004 mp	.00025 F
Vg mp F - 7	.007	.008	.0065	.003 Vg	.003 mp	.00025 F
Vg mp F - 10	.010	.011	.009	.003 Vg	.006 mp	.00025 F
Vg mp F - 15	.015	.0165	.014	.003 Vg	.009 mp	.00025 F
Vg mp F - 20	.020	.022	.019	.003 Vg	.010 mp	.001 F
Vg mp Vg - 10	.010	.0115	.009	.003 Vg	.003 mp	.003 Vg

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

Table I - Construction and thickness (Cont.)

Type	Thickness of composite			Thickness for each component 1/		
	Inch			Inch		
	Nominal	Maximum average	Minimum average	First layer suggested minimum	Second layer minimum average	Third layer suggested minimum
Vg mp Vg - 15	0.015	0.0165	0.014	0.003 Vg	0.004 mp	0.003 Vg
Vg mp Vg - 20	.020	.022	.019	.003 Vg	.009 mp	.003 Vg
Vg mp Vg - 25	.025	.0275	.024	.003 Vg	.014 mp	.003 Vg

1/ The thickness of mica paper (mp) shall equal or exceed the value shown. The first and third layer thicknesses are suggested minimums and can be varied. The plastic film (F) shall not exceed 0.001 inch except as so noted in this table.

Class 130		Class 155		Class 180	
Type	Form	Type	Form	Type	Form
F mp F - 5	S	F mp F - 5	S		
F mp F - 10	S				
F mp F - 15	S				
G mp F - 6	S&T	G mp F - 6	S&T	G mp F - 6	S&T
G mp F - 8	S&T	G mp F - 8	S&T	G mp F - 8	S&T
Vg mp F - 7	S	Vg mp F - 7	S	Vg mp F - 7	S
Vg mp F - 10	S	Vg mp F - 10	S		
Vg mp F - 15	S	Vg mp F - 15	S		
Vg mp F - 20	S	Vg mp F - 20	S		
Vg mp Vg - 10	S	Vg mp Vg - 10	S	Vg mp Vg - 10	S
Vg mp Vg - 15	S	Vg mp Vg - 15	S	Vg mp Vg - 15	S
Vg mp Vg - 20	S	Vg mp Vg - 20	S	Vg mp Vg - 20	S
				Vg mp Vg - 25	S

1.2.4 Designations.-

Numerals -- Thickness in thousands of an inch
F -- Plastic Film
mp -- Mica paper
G -- Glass Fiber Woven Fabric
Vg -- Varnished Glass Fiber Woven Fabric

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-271 - Paper, Wrapping, Waterproofed Kraft.
MMM-A-260 - Adhesive, Water-Resistant, (For Sealing Waterproofed Paper).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-576 - Box, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

FEDERAL (Cont'd)
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Set-Up
PPP-T-60 - Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-T-97 - Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U.S. Military Property.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D149 - Standard Methods of Test for Dielectric Breakdown Voltage and Dielectric Strength of Electrical Insulating Materials at Commercial Power Frequencies.
D257 - Standard Methods of Test for Electrical Resistance of Insulating Material.
D295 - Tentative Methods of Testing Varnished Cotton Fabrics and Varnished Cotton Fabric Tapes Used for Electrical Insulation.
D374 - Tentative Methods of Test for Thickness of Solid Electrical Insulation.
D618 - Standard Methods of Conditioning Plastics and Electrical Insulating Materials for Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.-

3.1.1 Mica paper.- The mica paper shall be composed of small mica particles reconstituted into flexible thin paper. The mica paper shall be clean, uniform, free of cracks and voids, and substantially free of conducting inclusions.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T08:38Z
Check the source to verify that this is the current version before use.

3.1.2 Plastic Film.- The plastic film shall be a polyester film. The 0.001 inch and 0.002 inch thickness shall conform to type G of MIL-I-631 and the 0.00025 inch thickness shall be electrical grade with a low electrical fault count.

3.1.3 Glass fiber woven fabric.- The glass fiber insulation shall be of glass fiber fabric having the warp and filling threads at right angles to each other.

3.1.4 Varnish glass fiber woven fabric.- The glass fiber insulation when varnished shall have a smooth, nontacky surface. A high quality baking varnish shall be used. The varnish or impregnant shall penetrate and adhere to the fabric, but shall not be corrosive to conductors after prolonged contact at elevated temperatures and high relative humidity. In the case of classes 130 and 155 insulation, the varnish shall be a high quality organic varnish free of any traces of silicones. In the case of class 180 insulation, the varnish shall be a high quality silicone varnish. The fabric before varnishing shall conform to 3.1.3.

3.1.5 Binder.- The binding material shall be a clear high quality insulating material uniformly distributed and used in the minimum amount needed to obtain the properties specified herein. The binding material shall provide an impregnating medium for the mica paper and also provide satisfactory adhesion between the mica paper and reinforcing layers as indicated by the absence of delamination or tearing through normal handling. The binder shall not dry out or "set-up" during normal storage (see 6.3) so as to result in stiffening of the insulation. For classes 130 and 155 insulation, the binder shall be a high quality organic resin free of any traces of silicones. For class 180 insulation, the binder shall be a high quality silicone resin.

3.2 Dimensions.-

3.2.1 Form S.- Unless otherwise specified, form S material shall be furnished trimmed and the length and width of the trimmed flat sheets shall be 36 inches plus or minus 1/2 inch. When sheet material is furnished as trimmed rolls the width tolerance shall be not greater than 1/16 inch.

3.2.2 Form T.- Unless otherwise specified, tape shall be furnished in rolls in widths of 1/2, 3/4, and 1 inch (standard sizes), within a tolerance of not greater than 1/32 inch, and a length of 36 yds. The tape shall not be spliced and not over two pieces shall be used in any one roll. The shorter of the two pieces shall be not less than 10 percent of the total length of the roll.

3.3 Performance characteristics.- The material shall conform to the applicable requirements of tables II and III.

Table II - Performance characteristics for classes 130 and 155.

Type	Dielectric strength - volts per mil (minimum average)				Volume resistivity Ohm-Cms (min. avg.)	Tensile strength lbs. per inch width (min. avg.)		Tear resistance lbs. (min. avg.)
	96/23/50 Straight	96/23/96 Straight	96/23/50 Bent length-wise	96/23/50 Bent cross-wise	96/23/96	96/23/50 Length-wise	96/23/50 Cross-wise	96/23/50 Both directions
F mp F - 5	1500	1200	1500	1500	10^{14}	30	30	8.0
F mp F - 10	950	700	950	950	10^{14}	30	30	12.0
F mp F - 15	900	650	900	900	10^{14}	90	90	20.0
G mp F - 6	800	500	800	800	-	60	25	
G mp F - 8	750	450	750	750	-	60	25	
Vg mp F - 7	1200	600	1100	1100	10^{13}	70	40	2.0
Vg mp F - 10	1100	550	1000	1000	10^{13}	70	40	3.0
Vg mp F - 15	1000	500	900	900	10^{13}	70	40	3.0
Vg mp F - 20	1100	500	1000	1000	10^{13}	70	40	5.0
Vg mp Vg - 10	1200	600	600	600	10^{8}	120	70	2.5
Vg mp Vg - 15	1100	550	600	600	10^{8}	120	70	4.0
Vg mp Vg - 20	1000	500	600	600	10^{8}	120	70	7.0

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

Table III - Performance characteristics for class 180.

Type	Dielectric strength - volts per mil (minimum average)				Volume resistivity Ohm-Cms (min. avg.)	Tensile strength lbs. per inch width (min. avg.)		Tear resistance lbs. (min. avg.)
	96/23/50 Straight	96/23/96 Straight	96/23/50 Bent length-wise	96/23/50 Bent cross-wise	96/23/96	96/23/50 Length-wise	96/23/50 Cross-wise	96/23/50 Both directions
G mp F - 6	700	425	500	500	-	60	25	-
G mp F - 8	650	400	500	500	-	60	25	-
Vg mp F - 7	1000	700	700	700	10^{13}	70	40	2.0
Vg mp Vg - 10	1200	700	850	850	10^{11}	120	70	3.5
Vg mp Vg - 15	1100	650	850	850	10^{11}	120	70	4.5
Vg mp Vg - 20	1000	600	850	850	10^{11}	120	70	5.0
Vg mp Vg - 25	950	575	750	750	10^{11}	120	70	5.5

3.4 Identification of individual sheets for stock.- Form S material furnished for stock shall be marked with the class and stock number in accordance with MIL-STD-130.

3.5 Workmanship.- The workmanship shall be first class in every respect. The finished product shall be reasonably free from wrinkles, creases, and other imperfections.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification: where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.-

4.2.1 Lot.- All material of the same class, form, type and size manufactured in one production run and offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.2.2 Sampling.- A random sample of sheets or rolls shall be selected from each lot offered for examination in accordance with table IV.

Table IV - Sampling for examination.

Lot size	Sample size	Acceptance number	Rejection number
Number of rolls of tape or sheets	Number of roll-specimens for each test	(test failures)	(test failures)
1-4	1	0	1
5-9	2	0	1
10-25	3	0	1
26-63	5	0	1
64-160	8	0	1
161-400	13	0	1
401-1,000	20	1	2
1,001-2,500	32	1	2
2,501-6,300	50	2	3
6,301-16,000	80	3	4
16,001-40,000	125	5	6

Source: https://assist.dla.mil -- Downloaded: 20260-19T08:38Z
Check the source to verify that this is the current version before use.

4.3 Quality conformance inspection and tests.-

4.3.1 Visual and dimensional examination.- Each of the sample sheets or rolls selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with this specification. Examination of the material of each sample roll shall be made on material remaining after several outer turns have been removed. Any sheets or rolls in the sample containing one or more visual or dimensional defects shall not be offered for delivery. If the number of defective sheets or rolls in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.3.2 Quality conformance tests.- Each of the sample sheets or rolls selected in accordance with 4.2.2 shall be tested to verify compliance with this specification.

4.3.3 Report of tests.- In addition to the copies required by the inspector, two copies of each test report shall be forwarded for information purposes via the inspector to the Naval Applied Science Laboratory (Code 9360) and the bureau or agency concerned.

4.4 Conditioning of test specimens.-

4.4.1 Nomenclature.- The designations indicating conditioning of test specimens shall be in accordance with ASTM D618.

4.4.2 Temperature and humidity tolerances.- Specified temperatures and relative humidities shall be maintained within the tolerances shown in Table V.

Table V - Tolerances.

Temperature		Relative humidity	
°C. 23	±°C. 2	% 50 96	±% 5 2

4.5 Test procedures.-

4.5.1 Thickness.- Specimen thickness shall be measured using a deadweight dial-type micrometer in accordance with method C of ASTM D374, except that the force exerted on the specimen shall be 20 ± 0.25 oz. and the anvil surface upon which the specimen rests shall be 2 inches in diameter. Readings shall be taken 10 seconds after lowering the presser foot. The thickness shall be determined as the average of 10 readings taken at intervals of approximately 6 inches along the length of each sample. The maximum, minimum, and average values shall be recorded.

4.5.2 Width and length - form S.- Measurements shall be made using a standard steel scale graduated to read to 1/64 inch. Five measurements each shall be made in the lengthwise and crosswise directions on the flat sheet samples. In the case of sample rolls, ten measurements at intervals of one foot shall be made of the width. The material shall lie flat on a smooth surface at the time of measurement. The average values shall be taken as the dimension of each sample.

4.5.3 Width and length - form T.- The length of the tape shall be determined by unwinding and measuring one of the sample rolls with a steel tape and then weighing it. The remainder of the sample rolls shall then be weighed and the weight compared with the first roll. Ten measurements at one foot intervals shall be made of the width using a steel scale and the average value shall be taken as the tape width.

4.5.4 Dielectric breakdown.-

4.5.4.1 Electrical apparatus.- Step up transformer, circuit breaker, voltage control and voltmeter shall be in accordance with ASTM D149. Power supply frequency shall not be greater than 100 cycles per second. The transformer shall have a rating of not less than 2 KVA, and the short time test at a rate of rise of 500 volts per second shall be used.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

4.5.4.2 Test electrode assembly.-

4.5.4.2.1 Straight specimens.- One-fourth inch diameter electrodes with edges rounded to 1/32 inch radius shall be used. To prevent flash over, 1/8 inch thick annular rubber or silicone rubber gaskets with 3/8-inch diameter center hole shall be employed around the electrodes. The testing device[1/] shall be arranged so that the pressure between the electrodes shall be approximately 6 inches.

4.5.4.2.2 Bent specimens.- Ninety-degree, V-shaped electrodes as shown on figure 1 shall be used. All edges of the electrodes shall be rounded to approximately 1/64-inch radius.

4.5.4.3 Bending jig.- A 90-degree, V-shaped jig shown on figure 2 shall be used.

4.5.4.4 Procedure.-

4.5.4.4.1 Straight.- Ten specimens, 4 inches by 4 inches, shall be conditioned 96/23/50 and ten conditioned 96/23/96. Dielectric breakdown shall then be measured on all specimens at 23°C.

4.5.4.4.2 Bent.- Ten specimens having a minimum size of 3 by 3-1/2 inches shall be cut and conditioned 96/23/50. Five specimens shall be bent in the lengthwise direction and five in the crosswise direction in bending jig applying a force of 10 pounds at a rate of 12 inches per minute (see 4.5.4.3). Specimens with dissimilar outer surface components shall be bent with the varnished or uncoated glass fabric in tension. Specimens shall then be placed in the V-shaped electrodes (see 4.5.4.2) and broken down at 23C. A force of 6 pounds shall be exerted by the upper electrode on the specimen.

4.5.4.5 Calculation of dielectric strength.- The dielectric strength in volts per mil shall be calculated by dividing the average dielectric breakdown by the average thickness as determined in 4.5.1.

4.5.5 Volume resistivity.-

4.5.5.1 Apparatus.- The apparatus shall be in accordance with ASTM D257 with a direct current potential of 100 to 200 volts applied to the specimen.

4.5.5.2 Test electrodes.- One electrode shall be of metal foil, 3 inches in diameter, one on surface of the specimen. The electrode on the other surface shall be of metal foil 2-5/8 inches in diameter. A concentric, metal foil guard ring electrode shall be used, and shall be 1/8 inch wide and spaced 1/16 inch from the inner electrode.

4.5.5.3 Procedure.- The electrodes shall be applied to three test specimens, 3-1/4 by 3-1/4 inches, by means of a very thin film of petrolatum, and shall be thoroughly rolled to exclude air. The electrodes shall be accurately centered with respect to each other. The volume resistance between the guarded and unguarded electrodes shall be measured at 23°C after voltage has been applied for one minute on each specimen after conditioning 96/23/96.

4.5.5.4 Calculation.- Volume resistivity, r, in ohm-centimeters, shall be calculated as follows:

$$r = \frac{RA \times 10^6}{t}$$

Where:

R = volume resistance in megohms.
A = area of guarded electrode in square centimeters.
t = average thickness of specimen in centimeters.

The average volume resistivity shall be reported.

1/ Such a device is described in the appendix of ASTM D295.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

4.5.6 Tensile strength.-

4.5.6.1 Apparatus.- Tensile strength shall be determined on a motor driven pendulum-type or constant rate of jaw separation testing machine.

4.5.6.2 Specimens.- The test specimens shall be cut using die I of method 4111 of FED-STD-601. The surface of the specimen shall be smooth and free from scratches. Five specimens cut lengthwise and five cut crosswise of the sheet shall be measured at 23°C.

4.5.6.3 Procedure.- The test specimens shall be placed in the testing machine and stretched until they break. The rate of separation of the jaws of the machine shall be 12 inches plus or minus 1/2 inches per minute. The tensile strength shall be reported in pounds per inch width.

4.5.7 Tear resistance (slot-cell).-

4.5.7.1 Apparatus.- The slot-cell tear resistance shall be determined using the tensile strength apparatus specified in 4.5.6.1 with specimens mounted in the slot-cell tear resistance jig shown on figure 3. The tearing edges which come in contact with the specimen have a radius of curvature of 1/32 inch and are set at an angle of 45 degrees.

Note: Working drawings of the jig are available from the U.S. Naval Applied Science Laboratory, Naval Base, Brooklyn, New York 11251.

4.5.7.2 Specimens.- Five specimens shall be torn in the tearing jig in the lengthwise direction and five in the crosswise direction at 23°C. after condition 96/23/50. The directions shall be reported as the directions in which the tears run. The specimens shall be 2-1/2 by 3 inches.

4.5.7.3 Procedure.- The specimens shall be inserted and aligned in the tearing jig. Specimens with dissimilar outer surface components shall be bent with the varnished glass in tension. The specimen holder shall be clamped to the upper jaw of the tensile machine. The tearing plate together with the attached guide plate shall then be clamped to the lower jaw. The maximum force in pounds developed in tearing through the edges of the specimen extending beyond the holder at a constant rate of 12 inches per minute shall be recorded on a chart. The average tear resistance in pounds for each direction of tear shall be calculated and reported.

4.6 Inspection of preparation for delivery.- Packaging, packing and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packing shall be level A or C, as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Form S insulation sheets shall be stacked in a like thickness and size in quantities specified (see 6.2) with a sheet of fiberboard the same size as the sheets, placed above and under the stack, secured in place with tape conforming to PPP-T-60. The stacks shall be individually wrapped with waterproof barrier material conforming to type C-1 or C-2 of UU-P-271 with all folds, seams, and laps sealed with water-resistant adhesive conforming to MMM-A-260.

5.1.1.2 Form S insulation rolls shall be individually wrapped with waterproof barrier material conforming to type C-1 or C-2 of UU-P-271. The wrapper shall be closed on the ends by means of inside and outside headers of the same material as the wrapper. Folds, laps, seams and headers shall be sealed with water-resistant adhesive conforming to MMM-A-260.

5.1.1.3 Form T insulation tape shall be individually packaged in accordance with method 1 A-8 of MIL-P-116. Packaged rolls of tape shall be individually packaged in folding or set-up boxes conforming to PPP-B-566 or PPP-B-676. Each box shall be closed and sealed in accordance with the box specification or appendix thereto.

Source: https://assist.dla.mil -- Downloaded: 202640-19T08:38Z
Check the source to verify that this is the currentversion before use.

5.1.1.4 Intermediate package.- Form S insulation rolls and form T insulation tape of like description and size packaged as specified in 5.1.1.2 and 5.1.1.3 shall be intermediate packaged in snug-fitting fiberboard boxes conforming to PPP-B-636 (class weather-resistant). Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints of the boxes are sealed with tape conforming to PPP-T-60 (type I or H, class 1). The box closures shall be as specified in the box specification or appendix thereto. The gross weight shall not exceed the weight limitation of the box specification.

5.1.2 Level C.- Form S and T insulation shall be packaged in accordance with the manufacturer's practice.

5.2 Packing.- Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.- The insulation, packaged as specified (see 6.2), shall be packed in containers conforming to anyone of the following specifications at the option of the contractor:

Specification	Classification
PPP-B-576	Class 2
PPP-B-585	Class 3 use
PPP-B-591	Class II
PPP-B-601	Overseas type
PPP-B-621	Class 2
PPP-B-636	Class weather-resistant

Shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2-inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes may be banded with tape conforming to type IV of PPP-T-97 and the appendix thereto. The gross weight of wood-cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiberboard boxes conforming to class weather-resistant of PPP-B-636, closed, sealed and banded as specified herein, and used as the shipping container need not be overpacked.

5.2.2 Level B.- The insulation, packaged as specified (see 6.2), shall be be packed in containers conforming to any one of the following specifications at the option of the contractor:

Specification	Classification
PPP-B-576	Class 1
PPP-B-585	Class 1 or 2 use
PPP-B-591	Class I
PPP-B-601	Domestic type
PPP-B-621	Class 1
PPP-B-636	Class domestic

Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiberboard boxes conforming to PPP-B-636, closed as specified herein, and used as the shipping container need not be overpacked.

5.2.3 Level C.- The insulation, packaged as specified (see 6.2), shall be packed in containers, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level in general shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Marking.- In addition to any special marking specified in the contract or order, or herein, each unit and intermediate package and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use.-

6.1.1 Class 130.- Class 130 material is used for generator, motor and transformer coils, slot liners and ground insulation, phase to phase insulation, phase and end insulation, primary secondary separation, coil exteriors, coil wrapper tapes, and similar uses. The highest temperature (limiting "hottest spot" temperature) to which class 130 insulation may be subjected to continuously with normal life-expectancy is 130°C.

6.1.2 Class 155.- Class 155 material is used for application as specified in 6.1.1 and where the highest temperature (limiting "hottest spot" temperature) is 155°C for normal life-expectancy.

6.1.3 Class 180.- Class 180 material is used for application as specified in 6.1.1 and where the highest temperature (limiting "hottest spot" temperature) is 180°C for normal life-expectancy.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, form, and type required (see 1.2).
(c) Nominal thickness (see table I).
(d) Form S material: Width, length and whether material is to be furnished trimmed (see 3.2.1).
(e) Form T material: Length of roll and width (see 3.2.2).
(f) Quantity (see 5.1.1.1).
(g) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Storage stability.- In ordering material under this specification, cognizance should be taken of its limited storage stability. Accordingly, purchasing activities, if possible, should avoid storage exceeding one year.

6.4 Certain provisions of this specification are the subject of international agreements (ABC-NAVY-STD-17). When amendment, revision or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Office so that appropriate action may be taken respecting the International agreement concerned.

6.5 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

International interest (see section 6)

Preparing activity:
Navy - SH
(Project 5970-N058Sh)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:38Z
Check the source to verify that this is the current version before use.

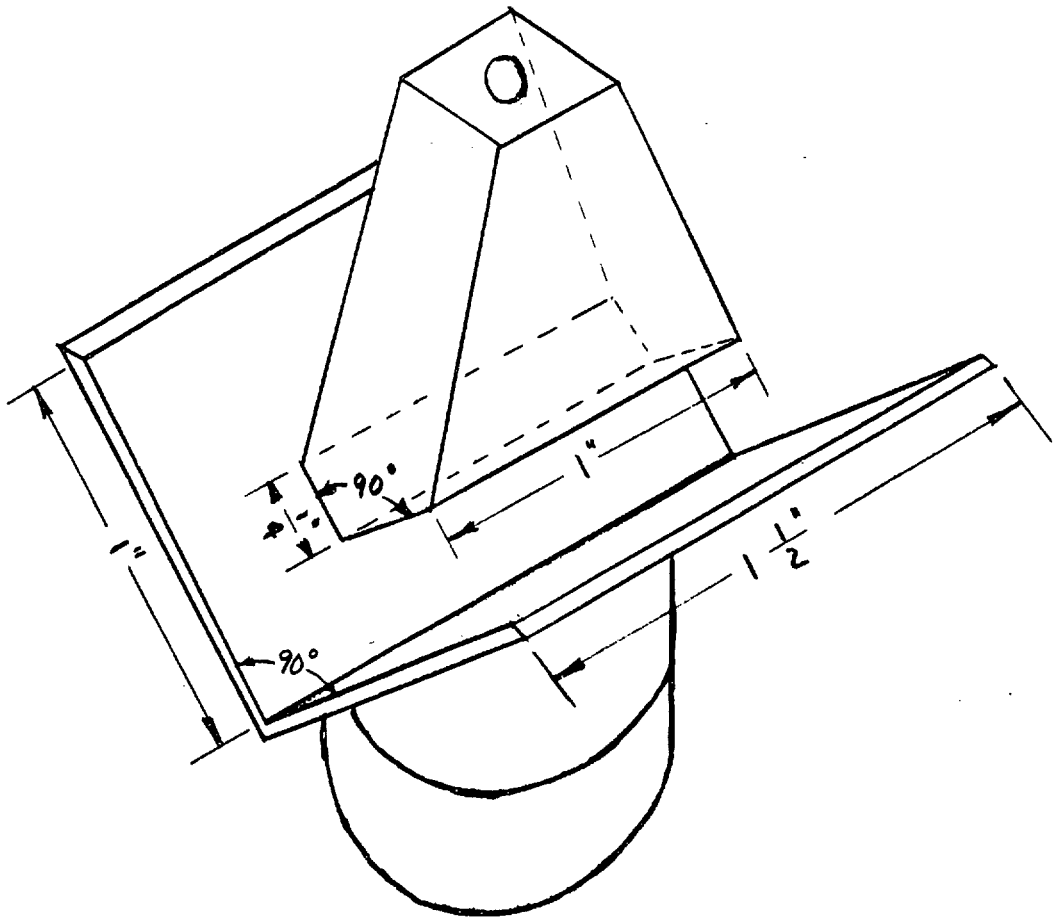


Figure 1 - Dielectric test electrodes 90° V-shaped.

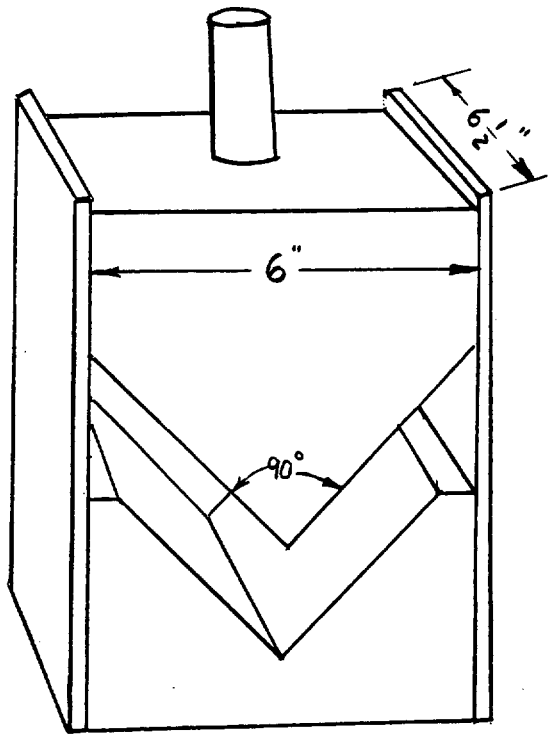


Figure 2 - Bending jig 90° V-shaped.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

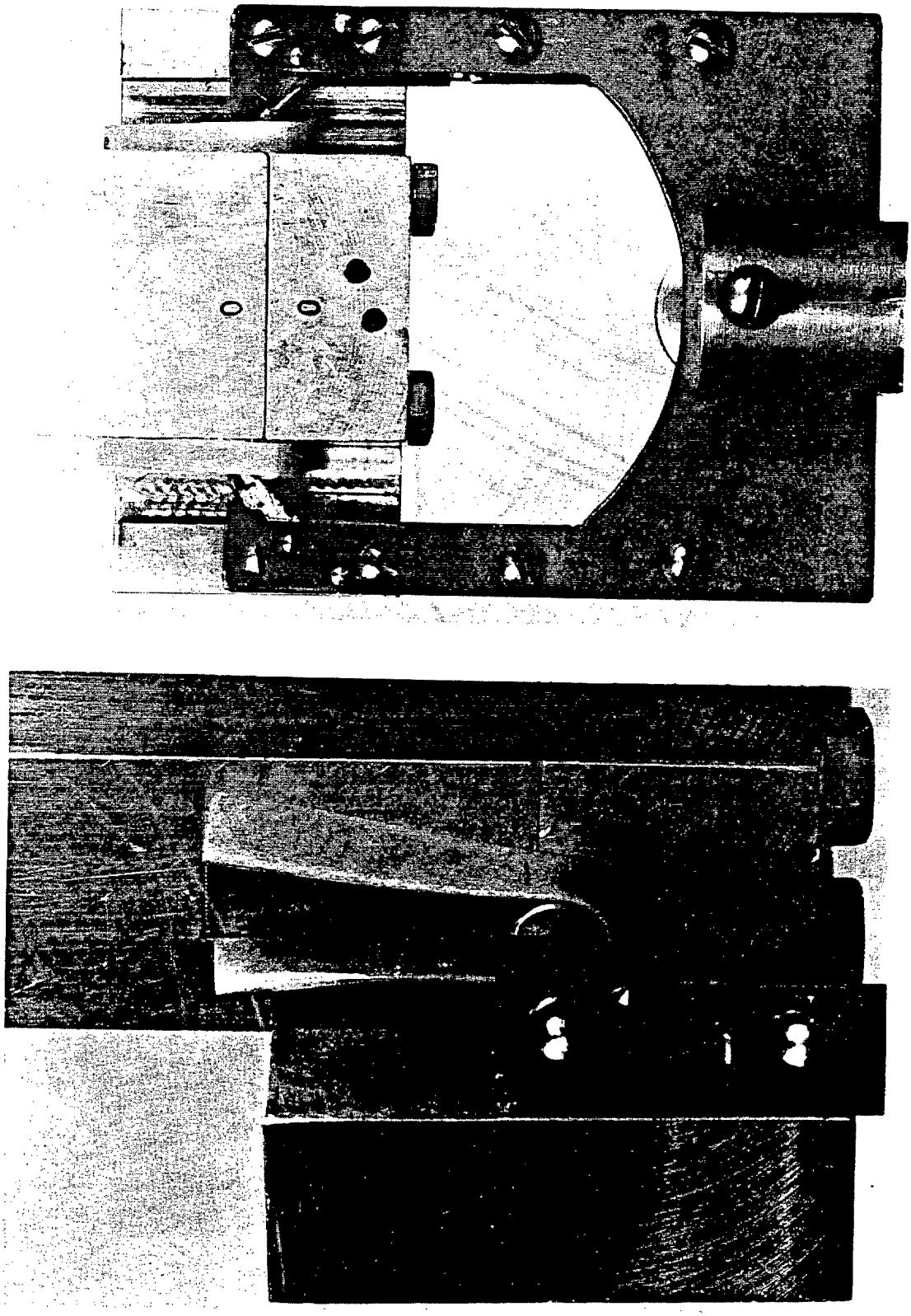


Figure 3 - NASL Slot-Cell Tear Resistance Jig Specimen in Position to Start Tear.

Source: https://assist.dla.mil -- Downloaded: 202610-19T08:38Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4865, WHICH IS OBSOLETE

D14000

Source: https://assist.dla.mil -- Downloaded: 2020-10-19T08:38Z
Check the source to verify that this is the current version before use.